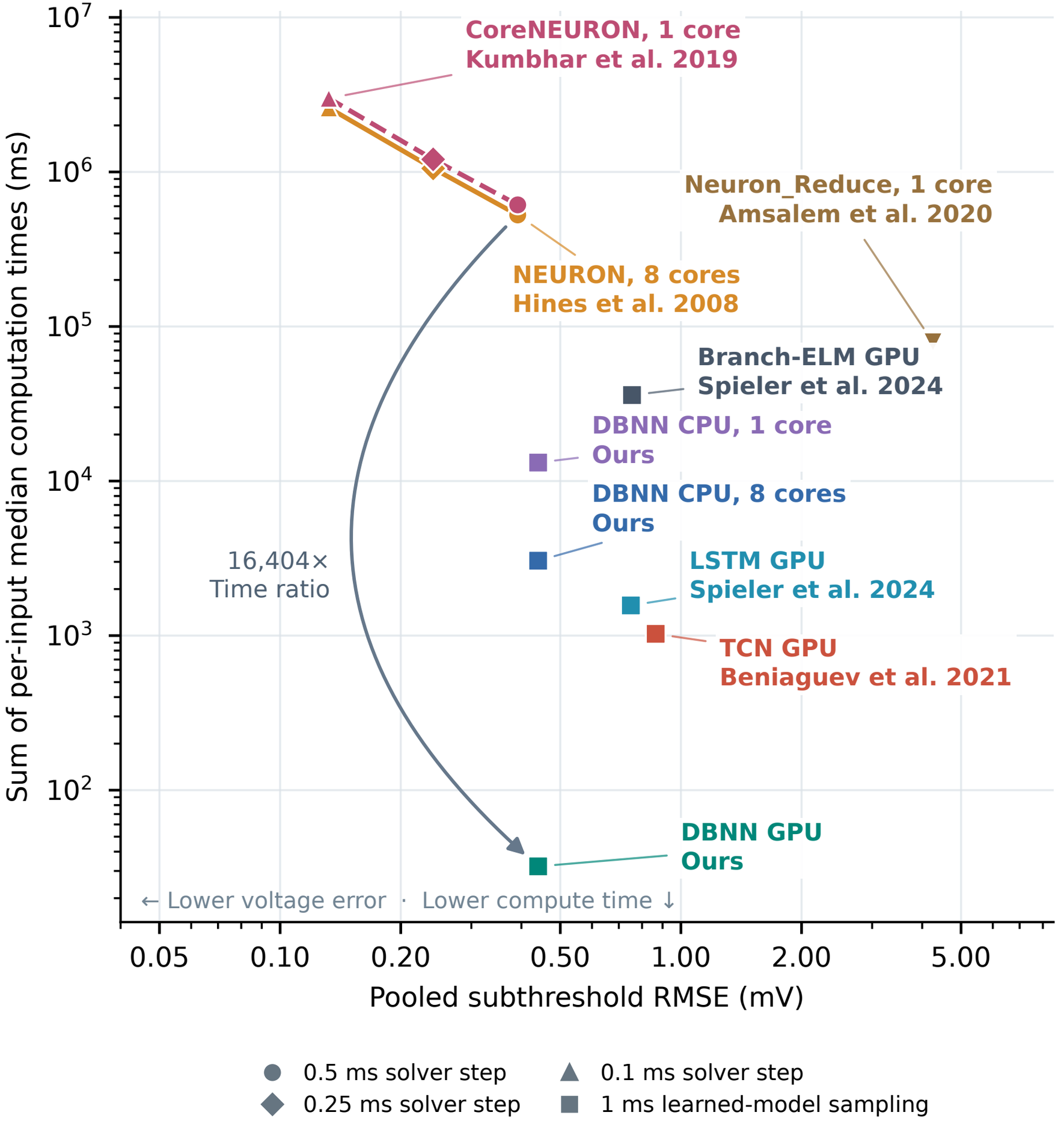
CoreNEURON, 1 core
Kumbhar et al. 2019
NEURON, 8 cores
Hines et al. 2008
Neuron_Reduce, 1 core
Amsalem et al. 2020
Branch-ELM GPU
Spieler et al. 2024
DBNN CPU, 1 core
Ours
DBNN CPU, 8 cores
Ours
LSTM GPU
Spieler et al. 2024
TCN GPU
Beniaguev et al. 2021
DBNN GPU
Ours
16,404×
Time ratio
← Lower voltage error · Lower compute time ↓
Sum of per-input median computation times (ms)
Pooled subthreshold RMSE (mV)
0.5 ms solver step
0.25 ms solver step
0.1 ms solver step
1 ms learned-model sampling